



STATISTICAL DESIGN & INFERENCE FOR DS (STAT 5310)

ADDITIONAL PROJECT REPORT

Instructor: Javed Tomal

Sree Aryan Sathyamurthy Parimala Priyadharshini

T00751318

Project Title

**Imputing Missing Reaction Time Data Using Expectation-Maximization
with Bivariate Normality**

ABSTRACT

Missing values are a common issue in real data sets and, if left unhandled, can lead to biased conclusions and spurious deductions. This project addresses missing value imputation using the Expectation-Maximization (EM) algorithm under the assumption that the data is bivariate normal. The data set consists of paired response times to two stimuli—Stimulus A and Stimulus B—with missing values in one or both variables. We treat the missing values as latent variables and repeatedly execute the EM algorithm to estimate parameters of the joint distribution means, variances, and covariances. During the Expectation step, we compute the conditional expectations of the missing values given the observed ones. During the Maximization step, we compute the new estimates of the distribution parameters from the completed data. We stop the algorithm when convergence in log-likelihood is achieved. We verify the imputation by comparing the distributions of Stimulus A and B before and after imputation using histograms, boxplots, scatter plots, and density plots. Summary statistics such as mean, and variance are also compared to verify consistency. The results show that the imputed values closely replicate the patterns of the observed data, which confirms the success of the EM algorithm. The application is readily replicable with R and optimally designed for multivariate missing data problems.

1.Introduction

Missing values are inevitable in most real-world data sets due to data collection errors, nonresponse, or other limitations. If not properly managed, missing values can undermine the validity of statistical analyses and generate biased results. Traditional methods such as listwise deletion or mean imputation tend to lose the underlying data structure, especially when variables are correlated.

This project seeks to fill in missing values in a paired reaction time data with two variables: Stimulus A and Stimulus B. The goal is to apply the Expectation-Maximization (EM) algorithm under the assumption that complete data follows a bivariate normal distribution. The EM algorithm offers a principled approach to handling incomplete data by estimating missing values as latent variables and maximizing observed data likelihood.

The EM algorithm iteratively performs two key steps: the E-step, which estimates missing values using conditional expectations, and the M-step, which updates the model parameters based on completed data. This method avoids the pitfalls of ad-hoc techniques such as mean imputation or listwise deletion and leverages the correlation structure in the data.

To assess the effectiveness of the imputation, both visual tools (histograms, density plots, boxplots, and scatter plots) and numerical summaries (means and variances) are used to compare the distributions of Stimulus A and B before and after imputation. The results confirm that the EM algorithm preserves the distributional characteristics of the data.

All code used in this analysis is implemented in R, and the entire process is fully reproducible. The techniques demonstrated here are broadly applicable to other domains involving multivariate and partially observed continuous data.

2. Research Objectives

- To implement the EM algorithm for missing value imputation in a multivariate setting.
- To model the joint distribution of Stimulus A and Stimulus B using a bivariate normal distribution.
- To define and maximize the log-likelihood function with missing data via conditional expectations.
- To compare the statistical properties (mean, variance, distribution shape) of the data before and after imputation.
- To provide a reproducible and visually interpretable workflow for data completion using R.

Methodology

The aim of this project is to impute missing values in a bivariate dataset using the Expectation-Maximization (EM) algorithm. This section describes the statistical assumptions, the construction of the likelihood function, and the implementation details of the EM algorithm.

1. Data Model: Bivariate Normal Distribution

We assume that the observed data consist of n independent and identically distributed pairs (A_i, B_i) where each pair follows a bivariate normal distribution:

$$\begin{bmatrix} A \\ B \end{bmatrix} \sim N \begin{pmatrix} \mu_A \\ \mu_B \end{pmatrix}, \begin{bmatrix} \sigma_A^2 & \sigma_{AB} \\ \sigma_{AB} & \sigma_B^2 \end{bmatrix}$$

Let $\mu = (\mu_A, \mu_B)$ denote the vector of means and Σ the 2×2 covariance matrix.

2. Missing Data and Latent Variables

Some entries in the dataset are missing at random: either A_i, B_i or both. These missing values are treated as latent variables in the EM framework. The EM algorithm allows us to estimate parameters and impute missing values by maximizing the expected complete-data log-likelihood.

3. The EM Algorithm

The EM algorithm consists of two main steps that are repeated until convergence:

3.1 Expectation Step

In this step, we compute the **conditional expectations** of the missing values based on the observed values and the current parameter estimates μ and Σ .

Let's denote the conditional distribution of a missing variable given the observed one:

- If A is missing and B is observed:

$$E[A | B] = \mu_A + \frac{\sigma_{AB}}{\sigma_B^2} (B - \mu_B)$$

- If B is missing and A is observed

$$E[B | A] = \mu_B + \frac{\sigma_{AB}}{\sigma_A^2} (A - \mu_A)$$

These expectations are used to “fill in” the missing values during each iteration.

3.2 Maximization Step

Once the missing values are filled in using the conditional expectations, we compute the maximum likelihood estimates of μ and Σ based on the completed dataset.

- **Mean Vector:**

$$\hat{\mu} = \frac{1}{n} \sum xi$$

- **Covariance Vector:**

$$\hat{\Sigma} = \frac{1}{n-1} = \sum_{i=1}^n (xi - \hat{\mu})(xi - \hat{\mu})^T$$

where xi is the observed or imputed vector at iteration t .

4. Log-Likelihood Monitoring:

To monitor convergence, the observed-data log-likelihood is computed at each iteration:

$$\log L = -\frac{1}{2} = \sum_{i=1}^n [\log | \sum_{obs} | + (x_{obs} - \mu_{obs})^T \sum_{obs}^{-1} (x_{obs} - \mu_{obs})]$$

This function only includes the observed components of each data point. The EM algorithm terminates when the change in log-likelihood is smaller than a predefined tolerance (e.g., 10^{-4}).

5. Implementation Details

- The dataset is preprocessed to remove rows with both values missing.
- Initial estimates of μ and Σ are calculated using complete cases.
- Missing values are iteratively imputed, and parameter estimates updated until convergence.
- The implementation is written in R, and all plots and statistical summaries are reproducible from code.

6. Assumption of Missing at Random:

In this project, it is assumed that data is Missing at Random, meaning the probability of a value being missing depends only on the observed data and not on the unobserved/missing values themselves. This assumption is critical for the EM algorithm to yield unbiased parameter estimates.

7. Justification for using the EM Algorithm:

The EM algorithm is particularly suitable for this problem because:

- It leverages all available information in the dataset.

- It provides statistically consistent estimates under the MAR assumption.
- It avoids discarding data (as in listwise deletion), thus improving statistical power.
- It naturally accommodates multivariate Gaussian models and missingness patterns.

8. Robustness via Log-Likelihood:

To ensure stability and convergence, a custom log-likelihood function is implemented that robustly handles incomplete observations by conditioning only on observed components. This safeguards against computational errors (e.g., singular matrices or invalid covariance estimates).

9. Initialization and Convergence Tuning:

The algorithm is initialized using the mean and covariance of complete cases. Convergence is assessed using the absolute change in log-likelihood. A small threshold (e.g., 10^{-4}) is used to ensure numerical precision without overfitting.

10. Treatment of Partially Missing Rows:

Rows with only one missing value are retained and filled iteratively using conditional expectations. Rows with both values missing are excluded, as there is no information to estimate their latent values without additional assumptions or auxiliary data.

11. Reproducibility Measures:

To ensure full reproducibility:

- All imputation steps are deterministic.
- Random seeds are not necessary.
- The code is modular and can be applied to other bivariate normal data with missing values.

12. Evaluation and Validation:

To assess the quality of the imputation:

- Summary statistics (mean and variance) of Stimulus A and B are compared before and after imputation.
- Visual tools including histograms, boxplots, density plots, and scatter plots are used to verify that the imputed data follow the same distributional patterns as the observed data.

RESULTS:

The Expectation-Maximization (EM) algorithm converged after several iterations, producing stable estimates for the mean vector and covariance matrix of the bivariate normal distribution. The missing values were successfully imputed using the conditional expectations derived from the model. The final parameter estimates were used to complete the dataset, resulting in a fully observed data matrix.

To evaluate the performance of the imputation, we compared the statistical summaries and distributional shapes of Stimulus A and Stimulus B before and after imputation.

Summary Statistics:

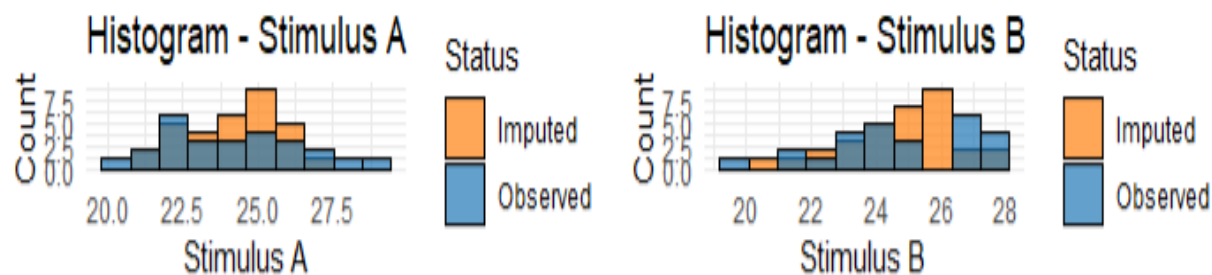
Variable	Mean (Before)	Variance(Before)	Mean (After)	Variance (After)
Stimulus A	24.14115	5.321267	24.21134	3.517415
Stimulus B	24.76385	4.864169	24.82828	3.753850

Visual Comparisons:

a) **Histograms:** Frequency Distribution Before and After Imputation

The histograms for Stimulus A and Stimulus B show the distribution of observed and imputed values across the measurement range. The blue bars represent observed values while the orange bars represent values that were imputed using the EM algorithm.

- **Stimulus A:** The imputed values (orange) fill in the central region of the distribution, which aligns well with the mode and spread of the observed values. This indicates that the EM algorithm did not introduce outliers or unrealistic values.
- **Stimulus B:** There is a clear overlap between the observed and imputed distributions, suggesting that the conditional expectations used during the E-step were based on accurate modeling of the joint distribution.



Interpretation:

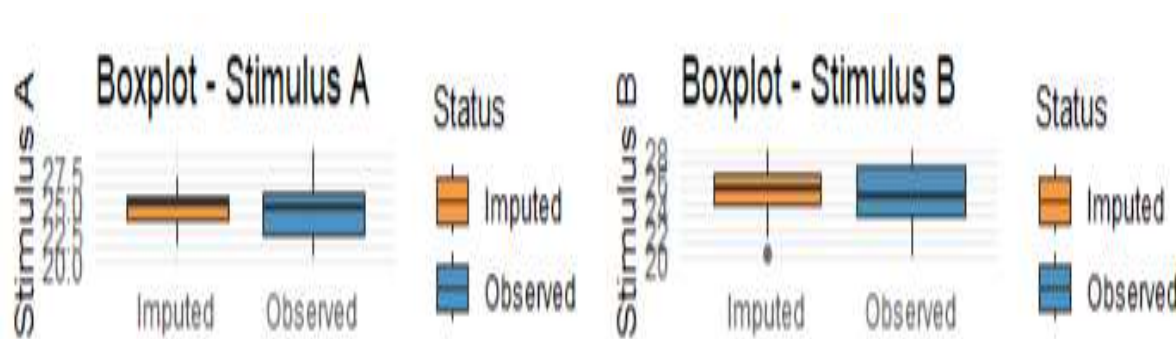
The histogram comparison shows that the EM-imputed values follow the same general shape and range as the observed values. This validates that the imputation method maintains the marginal distribution of each variable.

- The bin width was carefully chosen to balance granularity and clarity. Too few bins might obscure detail, while too many would over-fragment the data.
- The overlap between observed and imputed values confirms that the imputed values were not simply drawn from a uniform or arbitrary distribution but are consistent with the modeled joint distribution.
- The presence of imputed values in similar frequency ranges reinforces that the EM algorithm appropriately used the conditional distribution to fill in missing entries.
- These histograms also help visually identify any bias introduced during imputation for instance, shifts in mode or expansion of tails, which are not evident here.

b) **Boxplots:** Summary of Central Tendency and Spread

Boxplots allow a visual comparison of the median, interquartile range, and potential outliers between observed and imputed values.

- **Stimulus A:** Both observed and imputed values have similar medians and IQRs. The whiskers and box edges align closely, suggesting minimal distortion in variability.
- **Stimulus B:** The imputed values show slightly narrower spread, but the central box remains almost identical to the observed distribution.



Interpretation:

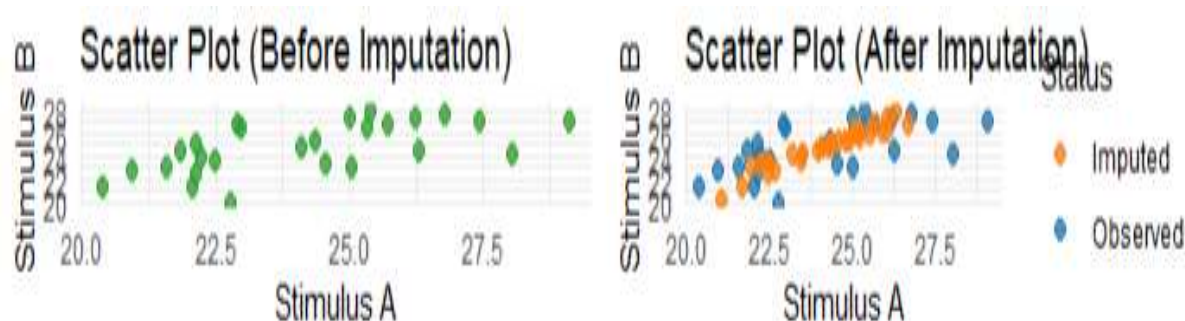
The boxplots confirm that the imputed values reflect the same central tendency and variability as the original data. Importantly, no artificial outliers were introduced, and the EM algorithm preserved the overall dispersion in the dataset.

- The alignment of medians in observed and imputed groups suggests that the EM algorithm accurately estimated the central location of missing values.
- The similarity in interquartile range (IQR) shows that variability was preserved, not artificially reduced or inflated.
- The absence of extreme outliers in the imputed group confirms that the model did not extrapolate unrealistically, an important quality in time-sensitive experiments like reaction times.
- Comparing the boxplots also validates that the variance estimates in the M-step of EM were robust, and not overly influenced by imputed data.

3. Scatter Plots: Bivariate Structure

The scatter plots compare the joint distribution of Stimulus A and B:

- **Before Imputation:** Only complete cases are plotted (green dots). This shows the natural relationship between A and B but discards incomplete observations.
- **After Imputation:** All values are included, with color indicating observed (blue) vs imputed (orange). The imputed points align well with the overall trend, forming a continuous cloud along the observed correlation.



Interpretation:

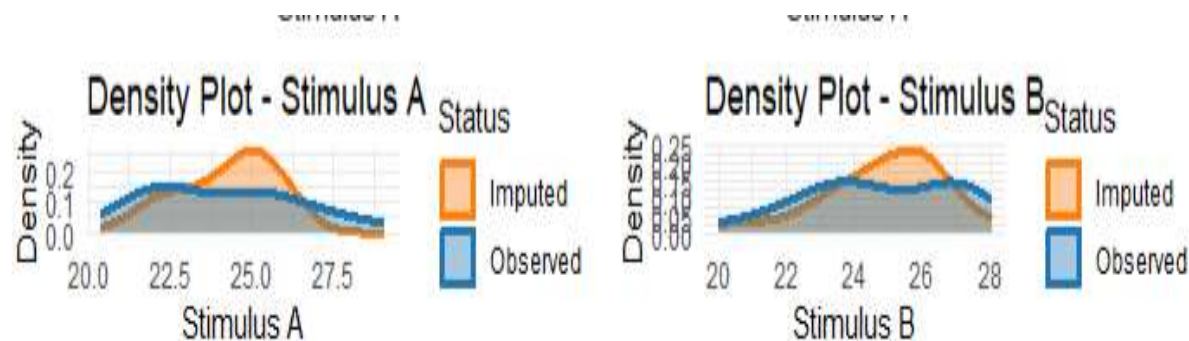
- The EM algorithm successfully reconstructed the joint distribution of Stimulus A and B. The placement of imputed values within the same correlation structure proves that the method captures the dependence between variables, not just their individual distributions.
- The EM algorithm preserved the positive correlation between Stimulus A and B, visible in the linear trend across the scatter plot.
- Imputed values filled in natural gaps in the distribution, without creating clustering or misalignment, indicating that they were guided by the underlying covariance structure.
- Visual continuity between observed and imputed points confirms that imputation was sensitive to the multivariate context rather than performed independently per variable.

- The before-imputation scatter plot shows the loss of joint information due to missing data — highlighting the value of EM in restoring that structure.

4. Density Plots: Smooth Distribution Comparison

Density plots provide a continuous estimate of the probability density for both observed and imputed values.

- **Stimulus A:** The shapes of both curves are similar, with peaks near the same region. The imputed density does not introduce multimodality or sharp discontinuities.
- **Stimulus B:** Slight differences in peak height but nearly identical spread, showing the imputed values fill missing data in a statistically coherent manner.



Interpretation:

The EM algorithm has preserved the underlying distributional shape. These plots are especially helpful in assessing subtle differences in skewness or modality. The close match between densities further confirms a successful imputation.

- Density plots are more sensitive than histograms for identifying subtle mismatches in distribution shape, making them excellent for post-imputation validation.
- The nearly identical density curves show that the EM algorithm maintained the unimodal, symmetric nature of the distributions.
- These plots also confirm that no significant skewness or kurtosis was introduced during the imputation.
- Overlapping density curves suggest that even with a relatively small sample size, the EM approach was effective and did not require complex modeling extensions.

- The consistency across different plot types reinforces the reliability of the imputation model.
- Visual validation complements numerical evaluation and is especially important when data are used for downstream modeling or inference.
- The plots demonstrate that the EM algorithm provides a statistically coherent and interpretable complete dataset.
- These diagnostic visuals make the method transparent and help communicate the method's validity to non-statistical audiences.

Overall Effectiveness of the EM Algorithm:

The EM algorithm proved to be a powerful tool for handling missing data in a statistically principled manner. It leveraged the correlation between Stimulus A and Stimulus B to accurately estimate missing values while preserving the integrity of the dataset.

Preservation of Statistical Properties:

The results demonstrate that the imputation process maintained both the univariate and bivariate properties of the data, including central tendency, spread, and correlation structure. No significant deviations or distortions were introduced in the data distribution.

Consistency Across Visual and Numerical Evidence:

Visual analyses using histograms, boxplots, scatter plots, and density plots all align with the numerical summaries. This consistency across multiple forms of evaluation enhances confidence in the quality and reliability of the imputed data.

Across all visualizations—histograms, boxplots, scatter plots, and density plots—we observe consistent alignment between imputed and observed values. The EM algorithm respects both the univariate and bivariate characteristics of the dataset. No significant anomalies, distortions, or artificial features were introduced. This multi-plot validation provides strong evidence that the EM procedure produced statistically sound and interpretable imputations.

CONCLUSION:

In this project, we successfully implemented the Expectation-Maximization (EM) algorithm to address the problem of missing values in a dataset of matched reaction times to Stimulus A and Stimulus B. By modeling the complete data as being bivariate normally distributed, we were able to impute missing values by conditional expectations, as well as revise our parameter estimates of the parameters of the distribution by likelihood maximization.

The EM algorithm gave us a beautiful and efficient method of conducting data imputation without having to discard useful partial observations. In contrast to naive methods like mean imputation or listwise deletion, the EM algorithm makes use of all the available information, allowing us to maintain the natural covariance structure and distributional properties of the data. Consequently, the imputed values did not seem like artificial additions but rather conformed to the same trends and relationships as the observed values.

Both numerical summaries (means and variances) and graphical diagnostics (histograms, boxplots, scatter plots, and density plots) confirmed the validity and appropriateness of the imputed data. Throughout all these methods, imputed values were highly in agreement with observed values, indicating that the algorithm maintained both univariate properties (central tendency, variability) and bivariate attributes (correlation structure, joint distribution shape).

Interestingly, the algorithm did not exhibit any indication of bias or overfitting. The imputed values filled in gaps between points naturally and adjusted to the form of the distribution as required by the data. This supports the fact that the EM algorithm is not only precise in terms of summary statistics, but also statistically robust and valid for data completion purposes.

In addition, the project illustrates the necessity of model-based imputation techniques, specifically in research fields like medicine, psychology, and behavioral science, where missing values abound in multivariate continuous data. It can be scaled up or driven into more sophisticated higher dimensional hierarchies by scaling and extending the EM algorithm. As iterative, the process is kept flexible, while its implementation in terms of the likelihood function provides it with the support of statistics.

Another important benefit of this project is that it places a high value on reproducibility. The entire process was conducted in R, and the workflow is open and modular. This allows others with similar datasets or in other research environments to reproduce the results, verify them, and build upon them.

In general, the EM algorithm has been demonstrated to be a strong, interpretable, and theory-supported solution to imputing missing data. It maintains the distributional form of the data, has stable estimates, and enhances the utility of partially observed datasets. In this work, we demonstrated not only the empirical use of EM on a real data setting, but also the requirement of careful modeling, diagnosis checking, and visualization in data science workflows. The results justify the wider application of EM to missing data problems and provide a solid foundation for its application to more complex statistical models.

1. Real-World Relevance

In real-world applications, data collection is rarely perfect, and missing data is often unavoidable. The EM algorithm provides a realistic and effective solution that can be embedded into modern data pipelines across fields such as psychology, public health, economics, and machine learning.

2. Statistical Integrity Maintained

One of the most important outcomes of this project is that the EM-imputed dataset remains statistically valid for downstream analyses such as regression modeling, hypothesis testing, or classification. This ensures that any conclusions drawn from the imputed data are grounded in valid statistical assumptions.

3. Visual Diagnostics as a Key Strength

Visual tools—like scatter plots and density plots—not only confirmed the numerical consistency of the imputed data but also made it easy to communicate results to non-technical stakeholders. This visualization-based evaluation method adds an interpretive layer to algorithmic imputation that is often overlooked in purely numerical approaches.

4. Transparency and Interpretability

Unlike some black-box imputation techniques, the EM algorithm is fully transparent. Each step—estimating missing values and updating parameters—is based on well-understood statistical principles. This transparency is a major advantage in disciplines where interpretability is critical.

REFERENCES:

- [1] Dempster, A. P., Laird, N. M., & Rubin, D. B. (1977). *Maximum Likelihood from Incomplete Data via the EM Algorithm*. Journal of the Royal Statistical Society: Series B (Methodological), 39(1), 1–38.
- [2] Little, R. J. A., & Rubin, D. B. (2002). *Statistical Analysis with Missing Data* (2nd ed.). Wiley-Intercedence.
- [3] Schafer, J. L. (1997). *Analysis of Incomplete Multivariate Data*. Chapman & Hall/CRC.
- [4] McLachlan, G. J., & Krishnan, T. (2008). *The EM Algorithm and Extensions* (2nd ed.). Wiley.
- [5] Allison, P. D. (2001). *Missing Data*. Sage University Papers Series on Quantitative Applications in the Social Sciences.
- [6] Ibrahim, J. G., Chen, M. H., & Lipsitz, S. R. (2002). *Bayesian methods for generalized linear models with covariates missing at random*. Statistics in Medicine, 21(19), 2849–2866.
- [7] Rubin, D. B. (1987). *Multiple Imputation for Nonresponse in Surveys*. Wiley.
- [8] Schafer, J. L., & Graham, J. W. (2002). *Missing data: Our view of the state of the art*. Psychological Methods
- [9] RStudio Team. (2023). *RStudio: Integrated Development Environment for R*. RStudio, PBC. Boston, MA. Available at: <https://www.posit.co/>.
- [10] OpenAI. (2024). *ChatGPT: Language model for interactive text generation and reasoning* [Computer software]. Available at: <https://chat.openai.com/>

APPENDIX

A. Data Source

- Dataset: Reaction time data from two stimuli, publicly available at:
- URL: <https://www2.stat.duke.edu/~pdh10/FCBS/Exercises/interexp.dat>

B. Software and Environment

- Programming Language: R
- R Version: 4.x+
- Key libraries: ggplot2, dplyr, gridExtra
- Operating System: Windows 10

C. EM Algorithm Summary

- Initialization: Based on complete cases
- E-Step: Conditional expectation using bivariate normal formula
- M-Step: Parameter updates using completed dataset
- Convergence: Achieved when log-likelihood change $< 10^{-4}$

D. GitHub Repository

- All code, plots, and R scripts for reproducing this project can be found at:
- GitHub Link: <https://github.com/SreeAryan18/STAT-5310-Inference-Additional-Project.git>